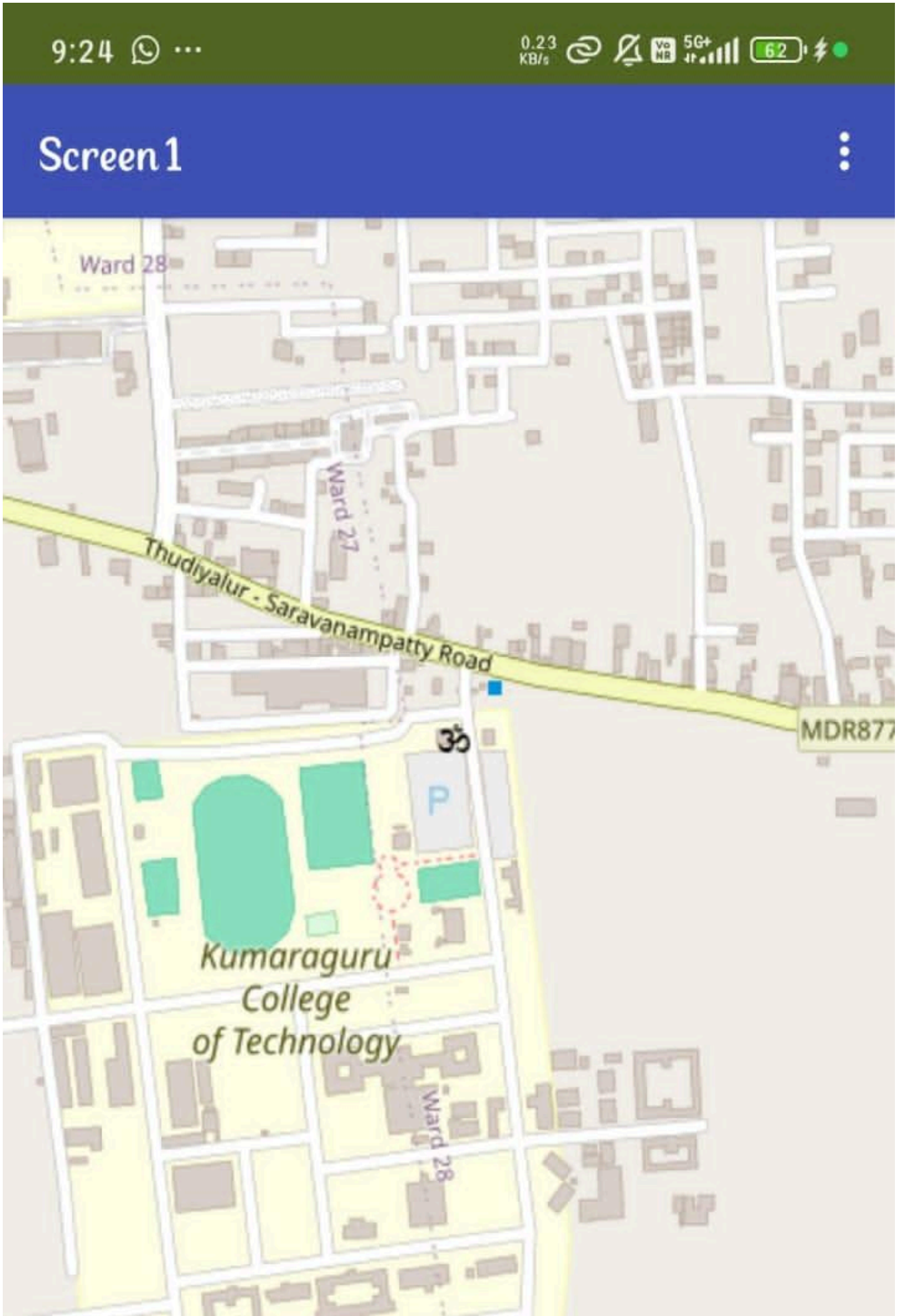
Location Tracker App – MIT App Inventor Project Explanation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0.23 KB/s
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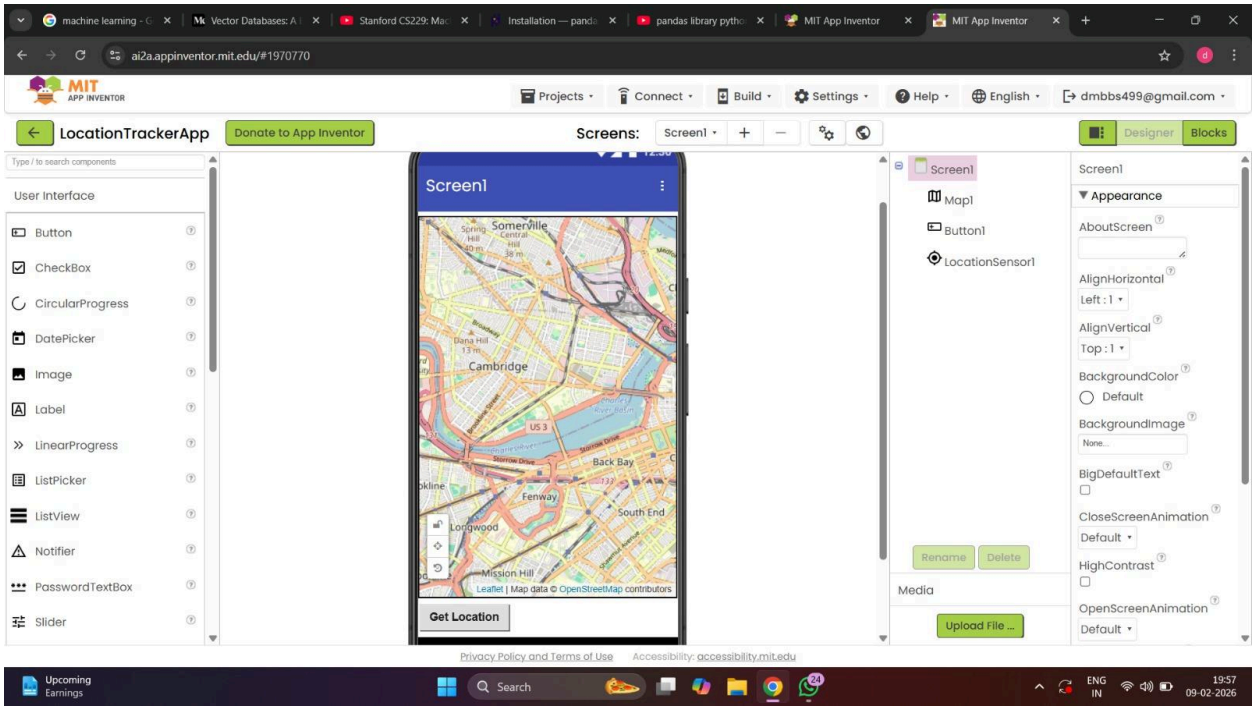
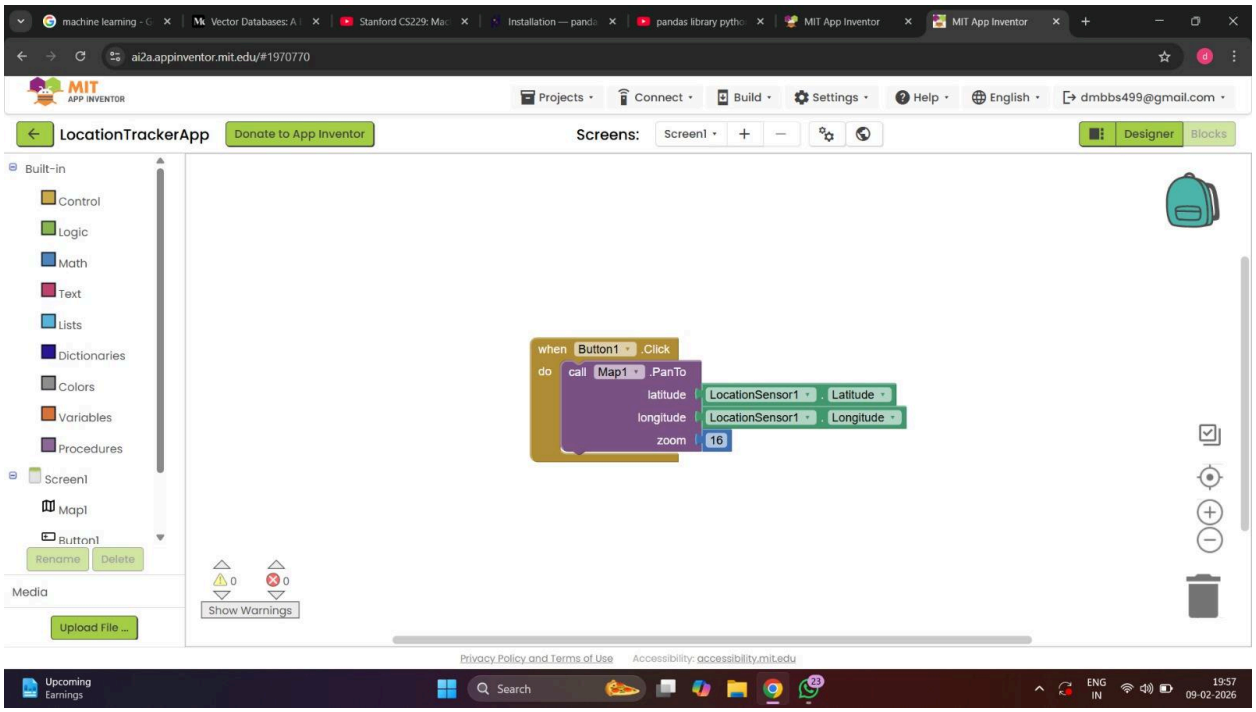
62
Screen 1
Ward 28
Ward 27
Thudiyalur - Saravanampatty Road
MDR877
P
Kumaraguru College of Technology
Ward 28





1. Introduction

This project is a **Location Tracker Mobile Application** developed using **MIT App Inventor**. The app uses the mobile phone's **GPS sensor** to detect the current location and display it on the map.

It demonstrates:

- GPS integration
- Map control
- Event handling
- Mobile app development without coding (block programming)

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2. Components Used

From your design screen, the following components are used:

🔹 Map1

- Displays OpenStreetMap on the screen.
- Shows the current location visually.

🔹 Button1 (Get Location)

- When clicked, it triggers the location update.

🔹 LocationSensor1

- Gets:
  - Latitude
  - Longitude
- Uses mobile GPS.

3. Block Code Explanation

From your block screen:

```
When Button1.Click
  Call Map1.PanTo
    latitude  = LocationSensor1.Latitude
    longitude = LocationSensor1.Longitude
    zoom      = 16
```

What This Means:

1. When user clicks **Get Location button**

2. The app reads:
    - Current Latitude
    - Current Longitude
3. The map automatically moves (PanTo) to that location.
4. Zoom level 16 shows a close street-level view.

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4. How the App Works

1. User opens the app.
2. Map loads.
3. User presses **Get Location**.
4. GPS reads coordinates.
5. Map centers to user's real-time location.
6. Current position is displayed.

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5. Concepts Used

✔ Event Handling (Button click event)
✔ GPS Sensor Integration
✔ Map Control Functions
✔ Latitude & Longitude usage
✔ Zoom Level Adjustment

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6. Real-Time Example (Your Screenshot)

Your screenshot shows:

- Location near **Kumaraguru College of Technology**
- Map centered on your campus area
- Blue marker showing current location

This confirms your app successfully detects real-world position.

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7. Applications of This App

This type of app can be used for:

- Live location tracking
- Delivery tracking systems
- Attendance tracking
- Navigation apps
- Safety apps

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8. Why This Project is Good

This project shows:

- Practical mobile development skill
- Real hardware integration (GPS)
- Understanding of maps and coordinates
- Basic logic and event-driven programming

It is a strong beginner-to-intermediate level mobile application.